# Uzbekistan OS
# Knowledge Schema Specification (KSS)
# Version 1.0

## 1. Purpose

Defines the canonical structure for every knowledge object in Uzbekistan OS, ensuring consistent storage, validation, retrieval, and AI consumption.

## 2. Knowledge Architecture

Official Source → Crawler → Raw Content → Parser → Structured JSON → Validation → Versioning → Chunking → Embeddings → Production.

## 3. Knowledge Hierarchy

Country → Domain → Topic → Document → Document Version → Chunks → Embeddings.

## 4. Universal Document Schema

Every knowledge object shares a common base schema containing metadata, content, relationships, citations, AI metadata, and version information.

## 5. Metadata

Includes title, slug, summary, keywords, audience, verification dates, effective dates, and publication status.

## 6. Versioning

Immutable document versions with major/minor/revision numbers, publication dates, superseded versions, and changelog.

## 7. Content Structure

Standardized sections including overview, requirements, steps, fees, processing times, exceptions, warnings, and FAQs.

## 8. Requirements & Steps

Structured requirement and workflow-step objects designed to power guided experiences and AI reasoning.

## 9. Fees & Processing

Reusable objects for costs, currencies, payment methods, processing times, and notes.

## 10. Relationships

Documents explicitly reference dependencies, related documents, workflows, and topics to support future knowledge graph traversal.

## 11. Citations

Every factual claim must reference one or more official sources with confidence scores and source metadata.

## 12. AI Metadata

Embedding model, token counts, chunk counts, retrieval priority, and quality score.

## 13. Domain Extensions

Specialized schemas for immigration, business, healthcare, tourism, and future domains while preserving the universal base schema.

# 14. Chunking

Semantic chunking based on headings, procedures, and complete concepts rather than arbitrary token boundaries.

# 15. Validation

Automated schema validation, relationship validation, citation verification, language checks, and publication requirements.

# 16. Knowledge Quality

Scoring based on source quality, completeness, freshness, structure, citations, translations, and human review.

# 17. Entity Types

Canonical entities such as countries, visas, permits, organizations, laws, workflows, offices, hospitals, banks, and contacts.

# 18. Knowledge Graph

Relationships connect entities (requires, issued_by, depends_on, applies_to, valid_for) to support multi-hop reasoning.

# 19. Definition of Done

All domains implement the universal schema, validation rules exist, chunking is standardized, citations enforced, and sample documents available.

# 20. Future Recommendation

Introduce a Canonical Entity Registry (CER) to provide stable identifiers for entities and simplify graph construction, retrieval, and long-term maintenance.